Annotating Vocal Responses (Hospital Testing)

From Internal CML Wiki

This is a page that focuses specifically on annotating verbal intracranial (and former RAM) tasks. The main difference between annotations for intracranial and scalp subjects are the many more sources of noise in intracranial subjects (doctors/nurses talking, beeping, participants usually have more vocalizations, etc.)

If you work in the CML Scalp EEG lab, go to the Annotating Vocal Responses (Scalp EEG) page

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Introduction

Some experiments require patients to make vocal responses. These vocal responses are recorded by the RAM testing laptop microphone and saved into sound files (.wav). Before these sessions can be processed and analyzed, a researcher must listen to the .wav files and *annotate* each word onset (thereby producing an .ann file). Accurately marking the word spoken and the word onset are both very important for ensuring our analyses are correct. Penn Total Recall (http://memory.psych.upenn.edu/TotalRecall) is a program developed by the lab to facilitate marking the identity and onset of vocal responses. Detailed instructions for annotating can be found on our Annotation Guide (https://memory.psych.upenn.edu/AnnotationGuide) wiki page.

Training: Annotation Gauntlet

The annotation gauntlet was designed to be used as a training tool for new annotators or as a means of double-checking the technique used by seasoned annotators. The annotation gauntlet is essentially a series of 12 lists of "patient" recalls with all the weird situations and exceptions to rules you are likely to encounter in the real world of RAM hospital testing.

An introduction and instructions for this are included in the following GitHub repository:
https://github.com/pennmem/annotation_gauntlet

Procedure

- Annotation is to be completed by the person that was in the room during testing
- Annotation should be on on the RAM testing laptop in which the patient session was run
- Annotate BEFORE uploading data to DCC

1. Open PennTotalRecall
    - PennTotalRecall should already be installed on all RAM testing laptops
    - Download PennTotalRecall (http://memory.psych.upenn.edu/TotalRecall) if it is not already installed on your account (search for it in spotlight first)
2. Select audio folder to be annotated
    - Click File -> Open Audio Folder
    - Choose session folder for the task to be annotated
        - Path will depend on where you choose to place the local version of the session folder, collected either from the testing computer if testing is done by internal lab member, or from a site's incoming directory (/data10/RAM/incoming)
3. Select wordpool corresponding to task to be annotated
    - Click File -> Select Wordpool
        - Path will again depend on where you place the local version of the wordpool. It is recommended to have backup versions of each wordpool for tasks currently being run in an accessible place on your local computer.
    - The bolded words are those that were presented to the patient during the task
    - Note: If annotating Spanish-language version of a task, choose the _noAcc wordpool version
        - Ex. RAM_wordpool_noAcc.txt or CatFR_WORDS_noAcc.txt
4. Mark vocalizations: See "Annotating" for details.
5. When finished with session, press “Mark Complete”
6. Upload annotations to rhino

Annotating

The goal of annotating is to mark the precise beginning of each vocal response you hear with an annotation mark that describes the response. For words, the annotation mark is just the recalled word (i.e. the participant says hat, the annotation mark is "HAT"). For other vocal sounds, like throat clearing or coughing, we use "<>" to mark it as a vocalization (the vocal sound isn't of interest, but we want to make a note to not look at that EEG signal). Rules for marking these annotations are described in the sections below.

See AnnotationGuide (http://memory.psych.upenn.edu/AnnotationGuide) for instructions on the keyboard commands for Penn Total Recall and detailed instructions on how to use TotalRecall to annotate sessions. You can also check out and edit the keyboard commands on Total Recall by clicking File > Edit Keyboard Shortcuts.

Completing a session

1. Each list needs to be annotated fully. After you've completed the list, double check in the annotations window that you haven't left any unannotated. All .wav files from that session should be "Marked Complete". Upload the data to the appropriate location in rhino.

Opening Previously Annotated Files

Sometimes you will need to open previously annotated files from another computer to edit your own work or check someone else's.

1. Navigate to the session directory
2. Open the folder as usual in Total Recall
3. Right click on the list(s) you want and hit "Continue Editing" before editing.
4. After annotating a session, press "mark complete" as you normally would. You changes should now be save in the .ann files

Verify_Annotation Script

For the integrity of our scientific analysis, it is important that our verbal experiments are annotated in a consistent way. That means that across annotators there should not be a great degree of variability. In order to cross check annotation reliability between different annotators, we use the verify_annotation script. This script allows two different annotators to annotate the same session and compare their performance to one another.

For this process, Annotator #1 annotates from the original session folder (ex. /data/eeg/R1124J/behavioral/FR1/**session_0**), while Annotator #2 annotates from a copy of the session folder that must be manually created (ex. /data/eeg/R1124J/behavioral/FR1/**session_0_QC**). If Annotator #2 tries to annotate from Annotator #1's same exact session folder (i.e. session_0), Annotator #2 will overwrite Annotator #1's annotations (aka .ann files). Once both annotators have annotated their respective session, the verify_annotation script can be run. This script will produce an output summarizing the comparable performance of the two annotators for that single session, taking into account such things as timing errors (when the annotators' disagreement on timing is outside the acceptable range [ex. + or - 25ms]) and mismatches (when the annotators disagree on what word was recalled [ex. one person annotates a word as DOCK and another annotates that same word as DUCK]). This script is compatible with RAM_FR, RAM_catFR, and RAM_PAL.

Click here for an example of QC_output.txt file (https://memory.psych.upenn.edu/InternalWiki/images/1/16/R1124J_FR3_0_QC_output.pdf)

Directions for cross checking annotation reliability:

1. Annotator #1 (**original annotator**) annotates the session as they normally would
2. Annotator #2 (**quality control annotator** or **QC annotator**) needs to create a new session folder to annotate from
    - **If annotating from rhino:**
        - Create a new folder in the task folder (ex. /data/eeg/R1124J/behavioral/**FR1**) with the session # and quality control ("QC") designation.
            - mkdir session_X_QC (ex. mk dir session_1_QC)
        - Copy the .lst and .wav files from the session_X folder you want to annotate into the folder you created above
            - cd session_X
            - cp *.lst /data/eeg/R1066P/behavioral/FR1/**session_0_QC** (substituting subject #, task, and session # of what you want to annotate)
            - cp *.wav /data/eeg/R1066P/behavioral/FR1/**session_0_QC** (same path as above)
            - *Note: all the .lst and .wav files should now be in your session_X_QC* folder
        - Annotate as you normally would, but from the session_X_QC folder
    - **If annotating from the RAM testing laptop:**
        - Create a new folder in the task folder (ex. /Users/exp/RAM/data/**FR1**) with the session # and quality control ("QC") designation.
            - ex. session_1_QC
        - Copy the .lst and .wav files from the session_X folder into the session_X_QC folder you just created
        - Annotate as you normally would, but from the session_X_QC folder
        - Use the RAM folder "upload_experiment" tool to upload Annotator #1's and Annotator #2's sessions to rhino
3. Once annotation is completed by both parties and uploaded to rhino, you can run the verify_annotation script **on rhino** (Note: both annotators must have the same number of .ann files/ aka fully have annotated the session)
    - Log on to rhino
    - From the task directory (/data/eeg/R1124J/behavioral/**FR1**) run the following script:
    - **verify_annotation <session_X> <session_X_QC> --hist >QC_output.txt**
    - ex. verify_annotation session_0 session_0_QC --hist >QC_output.txt
    - session_X = original session annotated (ex. session_1)
    - session_X_QC = second annotator
    - --hist = will produce two histogram plots showing the timing difference of the two annotators (#1 plot = timing difference for words & #2 plot = timing difference for vocalizations)
    - QC_output.txt = where the output of the script will be saved (if you run the script from the task folder, this output will be saved there)

To obtain the script, execute the following commands from rhino:

```
svn co $SVNROOT/misc/verify_annotation
cd verify_annotation
./setup
```

Calculating the AVERAGE cross checking reliability performance between two people (over multiple sessions)

1. Create a .txt file for Annotator #1, listing the paths to the annotation sessions
    - Example of Annotator #1's .txt file:
        - /data/eeg/R1066P/behavioral/FR1/session_2_SALLY
        - /data/eeg/R1118N/behavioral/FR1/session_0_SALLY
        - /data/eeg/R1073J/behavioral/FR1/session_4_SALLY
2. Create another .txt file for Annotator #2, listing the paths to the annotation sessions
    - Example of Annotator #2's .txt file:
        - /data/eeg/R1066P/behavioral/FR1/session_2_TIM
        - /data/eeg/R1118N/behavioral/FR1/session_0_TIM
        - /data/eeg/R1073J/behavioral/FR1/session_4_TIM
3. Run the verify_annotation script pointing to these newly created .txt files, instead of the individual annotations sessions

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Category: Hospital Testing

- This page was last modified on 15 March 2023, at 19:26.